Fiscal Year:	FY 2025	**Task Last Updated:**	FY 09/03/2024
PI Name:	Yamamoto, Namiko Ph.D.		
Project Title:	Micromechanics-Based Modeling of Cement Solidified in the Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nuy12@psu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617-417-1989
Organization Name:	Pennsylvania State University		
PI Address 1:	Aerospace Engineering		
PI Address 2:	556 White Course Drive		
PI Web Page:			
City:	University Park	**State:**	PA
Zip Code:	16802-1401	**Congressional District:**	12
Comments:			
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2020 Physical Sciences NNH20ZDA014N: Use of the NASA Physical Sciences Informatics System – Appendix G
Start Date:	11/01/2021	**End Date:**	10/31/2023
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Jackson, Enrique	**Contact Phone:**	
Contact Email:	enrique.m.jackson@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Postdoctoral associate Vishnu Saseendran, Ph.D., joined this project in May 2022, and his got promoted to Assistant Research Professor as of May 2023.		
COI Name (Institution):	Vishnu, Saseendran (Pennsylvania State University)		
Grant/Contract No.:	80NSSC22K0083		
Performance Goal No.:			
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Task Description:	The research goal of this proposal is to advance knowledge about the mechanical properties of cement binder fabricated in the microgravity environment using micromechanics-based modeling, based on the scanning electron microscopy (SEM) images and other materials data on the NASA Physical Science Informatics (PSI) database. As a crucial component to build shelters and habitats on the surface of the Moon and Mars using on-site resources, common cement binder formation in the microgravity environment was studied in the investigation titled as “Microgravity Investigation of Cement Solidification (MICS)” by Professor Aleksandra Radlinska’s group at Penn State [Ed. Note: see References]. Inspection of such cement, in comparison with the reference cement samples formed on the ground, indicated that the microgravity largely affected the microstructure formation: higher porosity, and larger or well-grown portlandite phase. The effect of this microstructure change on mechanical properties needs to be studied as these cements are aimed for structural applications, but such characterization is a challenge with these particular samples due to their sample porosity and small size. In this work, the Principal Investigator proposes to use the micromechanics analysis code with generalized method of cells (MAC/GMC) incorporated with the crack band model, to predict the structure-property relationship of these unique cement samples. The representative volume elements for the modeling will be created based on the SEM images from the NASA PSI database, and materials phases and associated mechanical properties will be assigned based on their X-ray powder diffraction (XRD) data also from the NASA PSI database. The broader impacts of this research proposal are 1) establishment of simulation processes to model mechanical and fracture behaviors of the cement samples from SEM images and 2) understanding about the effect of microgravity on microstructures and their mechanical properties of the cement. This study will provide the first reference simulation data for the cement binders fabricated in the microgravity environment. Such data can be compared with the currently limited experimental data to verify the structure-property trend and mechanisms about the cement samples prepared in microgravity. This study will also provide some guidance to update original ingredient mixtures and the hydration process (water-to-cement ratio, time, etc.) to improve mechanical properties of the cements formed in microgravity. Cement and concrete formation in space will be critical to form safe and economical habitats, and is expected to be in demand with the course of NASA’s Artemis program. In addition, the same or similar simulation process can be applied to study other cements and/or concrete materials formed, especially when those samples cannot be mechanically tested. References Moraes Neves, J., P.J. Collins, R.P. Wilkerson, R.N. Grugel, and A. Radlinska, "Microgravity effect on microstructural development of tri-calcium silicate (C3S) paste." Frontiers in Materials, Vol.6, No., 2019. pp. 83. Neves, J.M., "Advancing the Processing of Cement-based Materials in Extraterrestrial Environments," in Civil Engineering. 2020, The Pennsylvania State University: University Park, PA. Collins, P.J., "Microgravity's Influence on the Hydration of Tricalcium Aluminate and Gypsum Pastes," in Civil Engineering. 2019, The Pennsylvania State University: University Park, PA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The broader impacts of this study are 1) establishment of simulation processes to model mechanical and fracture behaviors of the cement samples from SEM images, and 2) understanding about the effect of microgravity on microstructures and their mechanical properties of the cement. This study will provide some guidance to update original ingredient mixtures and the hydration process (water-to-cement ratio, hydration time, etc.) to improve mechanical properties of the cements formed in microgravity. Cement and concrete formation in space will be critical to form safe and economical habitats, and is expected to be in demand with the course of NASA’s Artemis program. In addition, the same or similar simulation process can be applied to study other cements and/or concrete materials formed, especially when those samples cannot be mechanically tested.
	It is vital to close in the loop of process-structure-property relationship of cement hydrated in the microgravity environment, for successful fabrication and usage of cement in space. However, cement samples hydrated in space are limited by their sample size, thus rendering conventional experimental characterization techniques non-viable. In this work, the Principal Investigator (PI) used the micromechanics analysis code with high fidelity generalized method of cells, to predict the structure-property relationship of these unique cement samples. First, the sparse data of cement hydrated in microgravity environment was converted to high-fidelity 3D microstructure with the aid of a deep learning-based microstructure methodology. The reconstructed sub-volumes were compared against micro-CT data using statistical descriptors. Second, the sub-volumes (partitioned based on individual constituents) of such reconstructed 3D structures were fed into the NASA Multiscale Analysis Tool (NASMAT) to obtain homogenized properties of the cement. A semi-automated Python script was developed to complete the workflow. Owing to the high computational demand, the analyses were carried out in NASA Advanced Supercomputing (NAS) servers. This project provided the first reference simulation data for cement fabricated in the microgravity environment. In the previous years, the focus was on the first task. This year, the focus was on the second task on micromechanics-based prediction of basic mechanical properties. Originally, it was planned to directly import the Backscattered Electron (BSE) images of the hydrated samples as a user-defined element (2D analysis). Greyscale thresholding using the overflow method was showcased to estimate accurate porosity composition in both 1g and µg samples. Moreover, the identified thresholds may be used to generate the material card by assigning each pixel to a corresponding material based on the bounds. However, due to the high porosity content, 2D analysis is not viable (can also be attributed to the high water-to-cement (w/c) ratio). Therefore, it was decided to carry out a 3D analysis. The NASA Physical Sciences Informatics (PSI) database contains 2D BSE images; however, there are no 3D images which accurately capture morphology of both 1g and µg samples. Due to the lack of buoyancy, the CH crystals formed under microgravity conditions acquire the form of elongated platelets when compared against their 1g counterparts. To perform accurate 3D numerical studies, the Representative Volume Element (RVE) utilized must be representative of the actual microstructure morphology. In the field of image processing, Convolutional Neural Networks (CNN) is one of the prominent techniques of deep learning that have powerful learning capabilities. Recently, with advancements in machine learning, in particular deep learning, Gutierrez et al. proposed a CNN-based model to generate realistic solids textures of an arbitrary size. In general, this CNN-based synthesis model generates 3D microstructure by fitting statistical features between solid slices and 2D exemplars. This approach was adopted in this research to reconstruct C3S samples hydrated in both 1g and µg conditions. The CNN-based methodology was shown to yield promising results for cementitious materials, and is adopted here to reconstruct 3D microstructures of both 1g and µg samples. The algorithm was implemented in-house

Task Progress:	using Python on an Intel Xeon PC with 16 GB graphics processing unit (GPU) memory. Finally, the OpenGL platform coupled with PyVista were employed for visualization. First, a 250x250 pixels size window was chosen as BSE 2D exemplar. The rate of convergence of both samples seems to be consistent with convergence reached after ~1000 iterations; the only difference being the final converged values between 1g and µg sample that may be attributed to the inherent difference in statistical features between the samples. In general, from the perspective of microstructural image, cement paste is isotropic. The greyscale thresholding (overflow method in conjunction with local minima approach) was utilized to demarcate the bounds and assign material to a voxel in each RVE. The estimated porosity exhibited comparable values with the porosity values obtained through Mercury Intrusion Porosimetry (MIP): 48.4% and 69.4% for 1g and µg sample, respectively. The C-S-H phase composition was higher than the calculated values for reconstructed 1g RVEs. The C-H crystal composition in the reconstructed µg RVEs was also lower than that of 1g RVEs. Most importantly, the spatial orientation of CH crystals along with their randomness in orientation have been captured in the reconstructed 3D RVEs of 1g and µg samples. [Ed. Note: the PI notes that CH and C-S-H are hydration products.] The 3D 1g and µg microstructures reconstructed by CNN were assessed, by comparing them with the 3D virtual data obtained from micro-CT scans. Quantitative comparison was made by using the low-order probability distribution functions: two-point correlation function, two-point cluster function, and lineal-path function can be used to fully describe microstructure characteristics. For example, with the microgravity, the portlandite phase was observed to have an elongated plate-like morphology due to lack of gravity. Their clustering and connectivity can be described using these statistical descriptors. In addition, since these descriptors are still reduced order representation of micro-structures, qualitative visual comparison was also conducted. The portlandite phase is uniformly distributed with the 1g sample, while the portlandite phase shows the distinct plate-like morphology within the µg sample, in both CNN-reconstructed and microCT-inspected data. The reconstructed 3D volumes of both samples show similar microstructural characteristics to the 2D target image. After confirming the validity of the reconstructed sub-volumes, a workflow was created to seamlessly integrate such sub-volumes into NASMAT as RVEs, to be solved using the High-Fidelity Generalized Methods of Cells (HFGMC) theory. The challenge we faced has been the memory limitations to model the reconstructed RVE with the edge length of 276.4 µm (512 × 512 × 512 voxels, resolution 0.54 µm/pixel) using the NASMAT. Thus, the given volume was broken down into N sub-volumes, each comprising of m × m × m voxels. This strategy aided in overcoming the memory limitations and the elasticity matrix was calculated for a given 5123 volume of the 1g and µg samples. The samples were analyzed by discretizing Level-0 as 32 and Level-1 as 16 sub-cells. The input Young's modulus of the phases - portlandite, CH and C-S-H matrix were taken as 35 GPa, and 21.7 GPa, respectively. A very low Young's modulus value (~ 1 MPa) was considered for the porosity. The stiffness matrix was obtained from NASMAT and the decomposed Young's modulus, shear modulus, and Poisson's ratio were obtained. The Young's modulus of the 1g cement sample was found to be isotropic (~13.87 MPa), and consistent with E-modulus values reported in the literature for similar w/c ratio. For the first time, the Young's modulus of the cementitious system hydrated in microgravity (µg cement sample) was calculated to reduce from that of the 1g cement sample by 42%. In addition, a sensitivity study to understand the effect of length scales on 1g samples (2563, 1283 and 643 voxels, pixel resolution: 0.54 um/pixel) was conducted. Results indicate the convergence with respect to length-scale, provided the criterion for minimum edge-length of the RVE is met. Note that previously, we had established the minimum edge length required to select the optimum RVE size. This was based on the CH crystal length that varied ~ 100 – 115 µm, and an RVE of size ~ 2-2.5x must be selected.

Bibliography Type:	Description: (Last Updated: 09/26/2024)
Articles in Peer-reviewed Journals	Saseendran V, Yamamoto N, Collins PJ, Radlinska A, Mueller S, Jackson EM. "Unlocking the potential: Analyzing 3D microstructure of small-scale cement samples from space using deep learning." npj Microgravity. 2024 Jan 25;10(1):11. https://doi.org/10.1038/s41526-024-00349-9 , Jan-2024
Articles in Peer-reviewed Journals	Senthilnathan A, Saseendran V, Acar P, Yamamoto N, Sundararaghavan V. "Comparison and validation of stochastic microstructure characterization and reconstruction: Machine learning vs. deep learning methodologies" Acta Materialia. 2024 Oct 1;278:120220. https://doi.org/10.1016/j.actamat.2024.120220 , Oct-2024
Papers from Meeting Proceedings	Saseendran V, Yamamoto N, Kaleel I, Pineda EJ, Bednarcyk BA, Collins P, Radlinska A. "Multiscale modeling of reconstructed tricalcium silicate using NASA Multiscale Analysis Tool." 2024 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), Orlando, FL, January 8-12, 2024. Abstracts. 2024 AIAA Science and Technology Forum and Exposition (AIAA SciTech Forum), Orlando, FL, January 8-12, 2024. AIAA paper AIAA 2024-1001. https://doi.org/10.2514/6.2024-1001 , Jan-2024